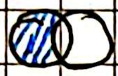
Discrete Mathematics — Sets

$\{\ \}$ → sets (not ordered)

$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{R}$	$\mathbb{Q}$	$\mathbb{C}$	$\varnothing$
normal	integer	real	fraction (rational)	complex	doesn't exist

$A \cup B$ → 

$A \cap B$ → 

$A \setminus B$ → 

$A \Delta B$ →  symmetrical difference

$A \times B$ → cartesian product

$\overline{A}$ → 

$A \subset B$ → proper subset of B (contain all)

doesn't contain all

$A \subseteq B$ → improper subset of B

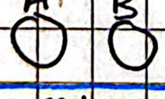
$c(A)$ → number of elements in A

Example: $C \Delta D = C \cup D \setminus C \cap D =$

$\{1,2,3,4,5,6\} \setminus \{3,4\} =$

$\boxed{\{1,2,5,6\}}$

$c(A) = m \quad c(B) = n \quad m \leq n$

operation			
$c(A \cup B)$	$m+n$	—	$c(B) = n$
$c(A \cap B)$	$c(\{\}) = 0$	—	$c(A) = m$
$c(A \setminus B)$	$c(A) = m$	—	$c(\{\}) = 0$
$c(B \setminus A)$	$c(B) = n$	—	$n - m$
$c(A \Delta B)$	$n+m$	—	$n - m$
$c(A \times B)$	$n \times m$	$n \times m$	$n \times m$

Lecture:

$\in \rightarrow$ notation $\quad \notin \rightarrow$ negation

Power Set:

$S = \{0,1,2,3\}$ so $2^S = \{\}$ or $\emptyset$ / $\{0,1,2,3\}$ / $\{0\}$ / $\{1\}$ / $\{2\}$ / $\{3\}$ / $\{0,1\}$ / $\{0,2\}$ / $\{0,3\}$ / $\{1,2\}$ / $\{1,3\}$ / $\{2,3\}$ / $\{0,1,2\}$ / $\{1,2,3\}$ / $\{0,1,3\}$ / $\{0,2,3\}$ 16 TOTAL

$c(S) \rightarrow 2^{c(S)}$ = total no.

$A \times B \rightarrow \{(a,c) : a \in A, b \in B\}$

$A = \{0,1,2\} \quad B = \{1,2\}$

$A \times B = \{(0,1), (0,2), (1,1), (1,2), (2,1), (2,2)\}$ 6 TOTAL

$c(A) \times c(B) = c(A \times B)$

$\max\{n,m\} \leqslant c(A \cup B) \leqslant n+m$

$0 \leqslant c(A \cap B) \leqslant \min\{m,n\}$

$0 \leqslant c(A \setminus B) \leqslant n$

if $c(A \setminus B) = 0 \rightarrow A \subset B$

$n-m$ or $m-n$ $\leqslant c(A \Delta B) \leqslant n+m$

$\overline{A \cup B} = \overline{A} \cap \overline{B}$

Quantifiers:

$\exists \rightarrow$ 'there exists' $\rightarrow$ existential quantifier

$\forall \rightarrow$ 'for all $\rightarrow$ universal quantifier

Functions

$x \mapsto f(x)$ domain → D range → R

So

$(x, f(x))$ & $x \in D$ & $f(x) \in R$

constant → $f(x) = c$ → straight line parallel to y

first order → $f(x) = mx + c$ → straight line

second order → $f(x) = ax^2 + bx + c$ → curve

factored form → $f(x) = a.\left(x - \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}\right)$. the opposite

polynomial →

exponential → $f(x) = a^x$ → $(a > 0, a \neq 1)$

logarithmic → $f(x) = \log_a x$ → $(a > 0, a \neq 1)$

trigonometric functions

absolute value function

sign function or signum function

1 in each row $f: D \to R, x \mapsto f(x)$

injective → $f(a) = f(b) \xrightarrow{\text{implies}} a = b$ aRc & bRc then a=b

surjective (right total) → for every element y in R there exists an $x \in D$ so $f(x) = y$ $\boxed{A = \{0,1\} \text{ \& } \text{rng}(R) = \{0,1\}}$

bijective → if it's injective & surjective mathematical induction:

1. Prove that it is true for $n=1$
2. Assume that it is done for $n=k$
3. Prove that it is true for $n=k+1$

Example: $1+3+5+\dots+(2n-1)=n^2$ x

num

$1+2+\dots+n = \frac{n(n+1)}{2}$ ✓

1. $n=1 \longrightarrow \frac{1(1+1)}{2}=1$
2. I assume if $n=k$ then $\frac{k(k+1)}{2}$
3. I prove if $n=k+1 \longrightarrow 1+2+\dots+k+(k+1) = \frac{(k+1)(k+2)}{2}$ so

$$\left(\frac{k(k+1)}{2}\right)+(k+1) \longrightarrow \frac{k(k+1)+2(k+1)}{2} = \boxed{\frac{(k+1)(k+2)}{2}}$$

Pascal's Triangle:

1

1 1

1 2 1

$(a+b)^3 \longrightarrow$ 1 3 3 1 $\rightarrow a^3+3a^2b+3ab^2+b^3$

1 4 6 4 1

$(a+b)^5 \longrightarrow$ 1 5 10 10 5 1 $\rightarrow a^5+5a^4b+10a^3b^2+10a^2b^3+5ab^4+b^5$

AN ORDERED PAIR $\rightarrow (a,b)$ normal brackets

Discrete Mathematics

Natural Numbers

Peano Axioms:

Axiom: Something we assume is true without proving it.

✓ For every natural number n, there exists uniquely a successor natural number.

✓ There is no natural number whose successor is 1.

Continue Defining ... presentation

Relatively Prime / Coprime Numbers

if $gcd(a,b) = 1$

greatest common divisor

Euclidean Algorithm:

$gcd(168, 45) =$

$168 = 45 \times 3 + 33$

$45 = 33 \times 1 + 12$

$33 = 12 \times 2 + 9$

$12 = 9 \times 1 + 3$

$9 = \boxed{3} \times 3 + 0$

3 is the gcd

Diophantine Equation:

$$ax + by = c \qquad \left[\begin{matrix} a, b, c \in \mathbb{Z} \\ x, y \in \mathbb{Z} \end{matrix}\right]$$

$\gcd(a,b) \mid c$

$$x = x_0 + t \times \frac{b}{\gcd(a,b)}$$

$$y = y_0 - t \times \frac{a}{\gcd(a,b)}$$

$$a\left(x_0 + t \times \frac{b}{\gcd(a,b)}\right) + b\left(y_0 - t \times \frac{a}{\gcd(a,b)}\right) = c$$

ex. $300x - 147y = 14$

$\gcd(300, 147) \mid c \rightarrow 3 \mid 14$? No

$300 = 147 \times 2 + 6$

$147 = 6 \times 24 + 3$

$6 = \boxed{3} \times 2 + 0$

Discrete Mathematics

Mathematical Induction

① $1^3 + 2^3 + \dots + n^3 = \left(\frac{n(n+1)}{2}\right)^2$

→ $n = 1$

LHS : $1^3 = 1$

RHS : $\left(\frac{1(1+1)}{2}\right)^2 = 1$

→ $n = k$

$1^3 + 2^3 + \dots + k^3 = \left(\frac{k(k+1)}{2}\right)^2$

→ $n = k+1$

$$\underbrace{1^3 + 2^3 + \dots + k^3}{} + (k+1)^3 = \boxed{\left(\frac{(k+1)(k+2)}{2}\right)^2}$$

$$\left(\frac{k(k+1)}{2}\right)^2 + (k+1)^3 =$$

$$= (k+1)^2\left(\frac{k^2}{2^2} + k + 1\right)$$

$$= (k+1)^2\left(\frac{k^2+4k+4}{2^2}\right)$$

$$= \frac{(k+1)^2(k+2)^2}{2^2}$$

$$= \boxed{\left(\frac{(k+1)(k+2)}{2}\right)^2}$$

Divisors & Divisibility

$A \rightarrow a_n a_{n-1} \dots a_2 a_1 a_0$

for 2 or 5 $\rightarrow$ if it divides a_0 then it's divisible

for ~~4~~ 4 or 25 $\rightarrow$ if it divides $a_1 a_0$ ✓

for 8 $\longrightarrow$ if it divides $a_2 a_1 a_0$ ✓

for 3 or 9 $\longrightarrow$ if $a_n + a_{n-1} + a_2 + a_1 + a_0$ ✓

for 11 $\longrightarrow$ if $a_0 \overset{\downarrow}{-} a_1 \overset{\downarrow}{+} a_2 \overset{\downarrow}{-} a_3 \dots$ ✓

① $10^{17} - 64 = 9999\dots36$

$36 \mid 10^{17} - 64$

$36 = 2^2 \times 3^2 = 4 \times 9$

divisibility by 4:

$4 \mid 36$ ✓

divisibility by 9:

$9 \mid 9 + 9 + 9 + \dots + 3 + 6$ ✓

$\boxed{\text{divisible by both}}$

because 4 & 9 are coprime (gcd = 1)

Discrete Mathematics

Diophantine Equations

ex. $180x + 378y = 36$

$\begin{bmatrix}0\\1\end{bmatrix}$ $\begin{bmatrix}1\\0\end{bmatrix}$ $\begin{bmatrix}-2\\1\end{bmatrix}$

$378 = 180 \times 2 + \boxed{18}$

$180 = \boxed{18} \times 10 + 0$

$180(-2) + 378(1) = 18$

$\times 2$

$180(-4) + 378(2) = 36$

Solutions
$$\begin{cases} x = -4 + t \times \frac{378}{18} = -4 + t \times 21 \\ y = 2 - t \times \frac{180}{18} = 2 - t \times 10 \end{cases}$$

Congruence $\rightarrow$ $a \equiv b \pmod{m}$

$m \mid (a-b)$

ex $\rightarrow$ $19 \equiv 10 \pmod{3}$

$3 \mid 19 - 10$ ✓

if $a \equiv b$ & $c \equiv d \pmod{m}$ then
$a \pm c \equiv b \pm d \pmod{m}$ & $a \times c \equiv b \times d \pmod{m}$

if $a \times c \equiv b \times c \pmod{m}$ & $\gcd(c, m) = 1$ then $a \equiv b$

Discrete Mathematics

Factorials

How many 0's are at the end of [100!]?

$10 = 2 \times 5$ bc we're checking no. of 0's

- Factors of 5 : $5^1, 5^2, 5^3 \rightarrow$ 125 above 100

$$= \left[\frac{100}{5^1}\right] + \left[\frac{100}{5^2}\right] + \left[\frac{100}{5^3}\right]$$

$$= 20 + 4 + 0 = \boxed{24}$$

$$100! = 5^{24} \times N \qquad N \in \mathbb{N}$$

- Factors of 2 : $2^1, 2^2, 2^3, 2^4, 2^5, 2^6, 2^7 \rightarrow 128$

$$= \left[\frac{100}{2^1}\right] + \left[\frac{100}{2^2}\right] + \left[\frac{100}{2^3}\right] + \left[\frac{100}{2^4}\right] + \left[\frac{100}{2^5}\right] + \left[\frac{100}{2^6}\right] + \left[\frac{100}{2^7}\right]$$

$$= 50 + 25 + 12 + 6 + 3 + 1 + 0 = \boxed{97}$$

$$100! = 2^{97} \times N \qquad N \in \mathbb{N}$$

$\longrightarrow$ $100! = 5^{24} \times 2^{97} \times N =$

$$(5 \times 2)^{24} \times 2^{73} \times N$$

$$10^{24} \times 2^{73} \times N$$

$\boxed{\text{24 ZEROs}}$

Euclidean Algorithm

$n = p \times q + r$ $\boxed{0 \leq r < q}$

→ $\gcd(a,b) \times \text{lcm}(a,b) = |a \times b|$

Extra Note = Odd = Even + odd only

1. $\gcd(360, 672) =$

$672 = 360 \times 1 + 312$

$360 = 312 \times 1 + 48$

$312 = 48 \times 6 + \boxed{24}$ → gcd

$48 = 24 \times 2 + 0$

→ 2 consecutive even numbers prove (product) $\boxed{8 \mid n}$

→ Product of 3 consecutive numbers prove $\boxed{3 \mid n}$

GCD & LCM : Find Canonical Representation

450	2
225	3
75	3
25	5
5	5
1	

420	2
210	2
105	3
35	5
7	7
1	

$450 = 2 \times 3^2 \times 5^2$

$420 = 2^2 \times 3 \times 5 \times 7$

gcd → take the lowest power

lcm → take the highest power

GCD → $2 \times 3 \times 5 = 30$

LCM → $2^2 \times 3^2 \times 5^2 \times 7 = 6300$

Total positive divisors :

$$1260 = 2^2 \times 3^2 \times 5 \times 7$$

$$3\left[\begin{array}{cc} 2^0 & 3^0 \\ 2^1 & 3^1 \\ 2^2 & 3^2 \end{array}\right. \quad 2\left[\begin{array}{cc} 5^0 & 7^0 \\ 5^1 & 7^1 \end{array}\right.$$

$$3 \times 3 \times 2 \times 2 = \boxed{36 \text{ positive divisors}}$$

✓ How many positive divisors of 7560 is coprime to 15?

$$7560 = 2^3 \times \cancel{3^3} \times \cancel{5} \times 7$$

$$15 = 3 \times 5 \rightarrow \text{disregard them}$$

$$(3+1) \times (1+1) = \boxed{8}$$

Discrete Mathematics

More Diophantine Equations

$70X + 80Y = 1420$

$7X + 8Y = 142$

$8 = 7 \times 1 + 1$

$\binom{0}{1} = \binom{1}{0} \times 1 + \binom{-1}{1}$

$(7 \times (-1) + 8 \times (1) = 1) \times 142$

$7 \times (-142) + 8 \times (142) = 142$

$X_0 = -142 \quad Y_0 = 142$

$X = -142 + 8t \qquad Y = 142 - 7t$

$\begin{cases} X \geq 0 \\ Y \geq 0 \end{cases} \rightarrow \begin{cases} -142 + 8t \geq 0 \\ 142 - 7t \geq 0 \end{cases}$

$\begin{cases} 8t \geq 142 \\ -7t \geq -142 \\ 7t \leq 142 \end{cases} \rightarrow \begin{cases} t \geq \frac{142}{8} \\ t \leq \frac{142}{7} \end{cases}$ Simplify (round) $\rightarrow \left\{ \boxed{18 \leq t \leq 20} \right.$

3 Possibilities

① $39^{28} \equiv ? \pmod{29}$

$\gcd(39,29)=1 \longrightarrow 39^{\varphi(29)} \equiv 1 \pmod{29}$

Since 29 is a prime number

$\varphi(29) = 29 - \boxed{1} = 28$

$39^{28} \equiv 1 \pmod{29}$

Note: When finding the last 2 digits do (mod 100)

② $23^{81} \equiv ? \pmod{50}$

$\gcd(23,50) = 1$

$23^{\varphi(50)} \equiv 1 \pmod{50}$

$50 = 2 \times 5^2$

$\varphi(50) = 50 \times \left(1-\frac{1}{2}\right) \times \left(1-\frac{1}{5}\right) = 20$

$(23^{20})^4 \equiv (1)^4 \pmod{50}$

$\downarrow$

$(23^{80} \equiv 1 \pmod{50}) \times 23$ (to make the power 81)

$$\boxed{23^{81} \equiv 23 \pmod{50}}$$

$54^{55^{56}} \longrightarrow 54^{55} \equiv a \pmod{13}$

$a^{56} \equiv ? \pmod{13}$

Discrete Mathematics

Complex Numbers

$i^2 = -1 \longrightarrow \sqrt{-1} = i$

① $x = \pm\sqrt{-4} \rightarrow \boxed{\pm 2i}$

② $x^2 + x + 1 \qquad \dfrac{-1 \pm \sqrt{-3}}{2}$

$$\underbrace{x_1 = \frac{-1}{2} - \frac{\sqrt{3}}{2}i \qquad x_2 = \frac{-1}{2} + \frac{\sqrt{3}}{2}i}_{\text{conjugates}}$$

$z = a + ib$

$\bar{z} = a - ib \rightarrow$ conjugate

$|z| = \sqrt{a^2 + b^2} \rightarrow$ absolute value

$|z| = \sqrt{z \times \bar{z}} \rightarrow$ can be written as.

$Re(z) = a \qquad Im(z) = b$

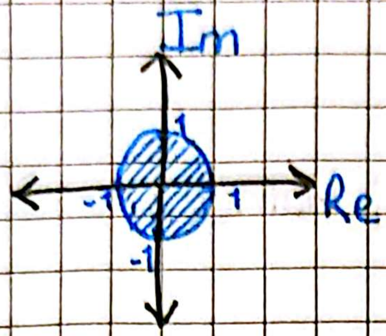
ex. $(2-i)^3 \longrightarrow 2^3 - 3 \times 2^2 \times i + 3 \times 2 \times i^2 + i^3$

$\boxed{-i^2 \times i = -1 \times i = i}$

$8 - 12i - 6 + i$

ex. $\dfrac{2-i}{(3-2i)(2+5i)} = \dfrac{2-i}{16+11i} \times \dfrac{16-11i}{16-11i} = \dfrac{32 - 22i - 16i + 11i^2}{16^2 + 11^2} =$

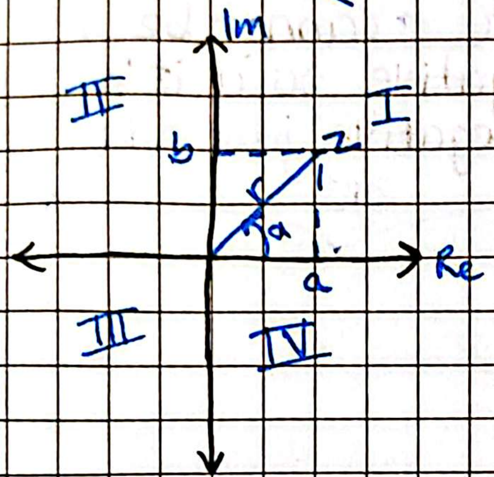
$$\boxed{\frac{21}{377} - \frac{38}{377}i}$$



$$|z| \leqslant 1 \rightarrow \sqrt{Im(z)^2 + Re(z)^2} \leqslant 1$$

circle equation

Trigonometric Form



$$Z = r(\cos\varphi + i\sin\varphi)$$

$$0 \leqslant \varphi \leqslant 2\pi$$

$$r = |z| = \sqrt{a^2 + b^2}$$

$$\tan\varphi_0 = \frac{|b|}{|a|} \; ; \; 0 < \varphi < \frac{\pi}{2}$$

$I \rightarrow \varphi = \varphi_0$	$III \rightarrow \varphi = \varphi_0 + \pi$
$II \rightarrow \varphi = \pi - \varphi_0$	$IV \rightarrow \varphi = 2\pi - \varphi_0$

ex. $1 - i$

$$r = \sqrt{1^2 + 1^2} = \boxed{\sqrt{2}}$$

$$\tan\varphi_0 = \frac{|-1|}{|1|} = 1 \rightarrow \tan^{-1}(1) = \varphi_0 = \boxed{\frac{\pi}{4}}$$

Quadrent IV so $\rightarrow \varphi = 2\pi - \varphi$

$$= 2\pi - \frac{\pi}{4} = \boxed{\frac{7\pi}{4}}$$

$$\boxed{\sqrt{2}\left(\cos\left(\frac{7\pi}{4}\right) + i\sin\left(\frac{7\pi}{4}\right)\right)}$$

CamScanner

Exponential Form

$\rightarrow z_1 = r_1 e^{i\varphi_1}$; $e^{i\varphi_1} = \cos(\varphi_1) + i\sin(\varphi_1)$

$z_2 = r_2 e^{i\varphi_2}$

$z_1 z_2 = r_1 r_2 e^{i(\varphi_1 + \varphi_2)}$

$\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\varphi_1 - \varphi_2)}$

$\frac{1}{z_1} = \frac{1}{r_1} e^{-i\varphi_1}$

$(z_1)^n = r_1^n e^{i \times n \times \varphi_1}$

$\varphi \rightarrow$ cannot be negative so if it's negative just add 2π

N^{th} Root

of $z = r(\cos\varphi + i\sin\varphi)$

$X_0 \rightarrow X_1$ add the value of the angle in between

$X^n = z$

$$X_k = \sqrt[n]{r}\left(\cos\left(\frac{\varphi + 2k\pi}{n}\right) + i\sin\left(\frac{\varphi + 2k\pi}{n}\right)\right)$$

$k = \{0, 1, 2, \ldots, n-1\}$

The angle between X_k & $X_{k+1} \rightarrow \frac{2\pi}{n}$

When finding the 2nd root $\Rightarrow n = 2$ $k = \{0, 1\}$
Solve X_0 & X_1

When finding the 3rd root $n = 3$ $k = \{0, 1, 2\}$
Solve X_0 & X_1 & X_2

Example →

$n = 3 \quad k = \{0, 1, 2\}$

$X_0 = \sqrt[3]{81}\left(\cos\left(\frac{\pi/5 + 0}{3}\right) + i\sin(-)\right) =$

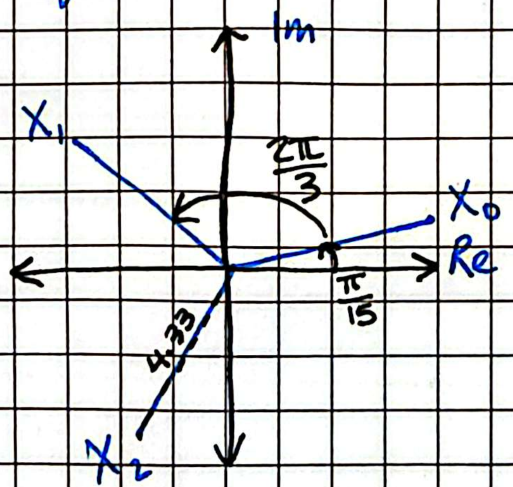
$4.33\left(\cos\left(\frac{\pi}{15}\right) + i\sin\left(\frac{\pi}{15}\right)\right)$

Angle Between $= \frac{2\pi}{n} = \boxed{\frac{2\pi}{3}}$

$X_1 = 4.33\left(\cos\left(\frac{\pi}{15} + \frac{2\pi}{3}\right) + i\sin(-)\right) =$

$4.33\left(\cos\left(\frac{11\pi}{15}\right) + i\sin\left(\frac{11\pi}{15}\right)\right)$

Repeat for X_2



Discrete Mathematics

Polynomials

3rd order term → ax^3

If $z \in \mathbb{C}$ is a root of P then
→ $\bar{z}$ is also a root of P.

Note: Minus the roots from the x $(x-3)(x-(-2))$

$n = p \times q + r \quad : \quad 0 \leq r < q$

$P(X) = Q(X) \times L(X) + R(X)$

$0 \leq \deg(R) < \deg(L)$

$ax^2 + bx + c = 0$

1. if $a+b+c=0$ $\quad X_1 = 1 \quad X_2 = \frac{c}{a}$
2. if $a-b+c=0$ $\quad X_1 = -1 \quad X_2 = \frac{-c}{a}$
3. $X_{1,2} \quad \frac{-b \pm \sqrt{b^2-4ac}}{2a}$

ex. $x^3 + 3x^2 + 3x + 1 \rightarrow (x+1)^3$

ex. $x^3 - 6x^2 + 12x - 8 \rightarrow (x-2)^2$

multiplicity 2 means → to the power of 2

Horner's Method

	x^5	x^4	x^3	x^2	x^1	x^0
+	2	-3	1	-5	2	-4
2	2	1	3	1	4	$\boxed{4}$
x	x^4	x^3	x^2	x^1	x^0	

$\boxed{4} \rightarrow P(2)$

$\boxed{4} \rightarrow$ Remainder when $\dfrac{P(x)}{x-2}$

SO $\rightarrow 2x^5 - 3x^4 + x^3 - 5x^2 + 2x - 4 =$

$$(x-2)(2x^4 + x^3 + 3x^2 + x + 4)\ \boxed{+4}$$

remainder

DIVISIONS

① $(-3x^3 + 48x) \div (x-4)$

	x^3	x^2	x	x^0
	-3	0	48	0
4	-3	-12	0	0

$\longrightarrow (x-4)(-3x^2 - 12x)$

Euclidean Division

divide both big degrees

$$\begin{array}{r|l} x^4 + x^3 - 3x^2 - 4x - 1 & x^3 + x^2 - x - 1 \\ \hline -x^4 - x^3 + x^2 + x & x \\ \hline 0 + 0 - 2x^2 - 3x - 1 & \end{array}$$

$$= \underbrace{(x^3 + x^2 - x - 1)(x)}_{\text{what we calculated}} \underbrace{-2x^2 - 3x - 1}_{\text{remainder}}$$

Discrete Mathematics

Combinatorics

Permutation → $x!$

Permutation of multi-sets → $P_n^{l_1 \ldots l_k} = \frac{n!}{l_1! \ldots l_k!}$

Ex. $P_{15}^{3,5,7} = \frac{15!}{3! \times 5! \times 7!} = \binom{15}{3} \times \binom{12}{5} \times \binom{7}{7}$ $\boxed{\binom{n}{k} = \frac{n!}{k! \times (n-k)!}}$

$\frac{15!}{3! \times \cancel{12!}} \times \frac{\cancel{12!}}{5! \times 7!} \times 1 = \boxed{\frac{15!}{3! \times 5! \times 7!}}$

Roundtable : $(n-1)!$

Permutation with repitition : $P(n,k)^{rep} = n^k$

Combination with repitition : $C(n,k)^{rep} = \left(\binom{n}{k}\right) = \binom{n+k-1}{k}$

2 Vector Formulas :

[To prove vector independence] : $\lambda_1 V_1 + \lambda_2 V_2 = 0$

[To prove with respect to a base] : $\lambda_1 V_1 + \lambda_2 V_2 = b$ base

To make equal to 0 → $L - \lambda P$ pivot line

Systems of Linear Equations

Rank(A) < rank(A/b) → no solution/inconsistent

- b is not in the subspace generated by the columns of A
- det(A) = 0
- Rows/columns of A are linearly dependent

Rank(A) = Rank(A/b) → solvable

- b is in the subspace generated by the columns of A.

$$\begin{array}{c} A \\ \text{Rank of } A \left\{ \begin{pmatrix} \bullet & \bullet & | & \bullet \\ \bullet & \bullet & | & \bullet \end{pmatrix} \right. \end{array}$$

$$\text{Rank of } A|b \left\{ \begin{pmatrix} \bullet & \bullet & | & \bullet \\ \bullet & \bullet & | & \bullet \end{pmatrix} \right\}$$

Rank(A) = # of col. of A : (determined):

- 1 unique solution
- det(A) ≠ 0
- Columns/rows of A are linearly independent

Rank(A) < # col. of A : (undetermined):

- ∞ many solutions
- det(A) = 0
- columns/rows of A are linearly dependent

$$\boxed{\left(\begin{array}{ccc|c} -1 & 2 & 4 & 5 \\ 0 & 0 & 0 & 9 \\ 0 & 0 & 0 & 7 \end{array}\right) \begin{array}{l} \rightarrow \text{Rank}(A) = 1 \\ \rightarrow \text{Rank}(A|b) = 3 \end{array}}$$

Discrete Mathematics

Matrices

Det :

- Switching Lines → x -1
- Zero's under diagonal → multiply diagonal

ex. $\begin{vmatrix} 3 & -1 & 2 \\ 0 & 0 & 5 \\ 0 & 1 & -5 \end{vmatrix}$ $[L_2 \leftrightarrow L_3]$

x -1

$- \begin{vmatrix} 3 & -1 & 2 \\ 0 & 1 & -5 \\ 0 & 0 & 5 \end{vmatrix}$ $[3 \times 1 \times 5 \times -1 = -15]$

(Det = -15)

Inverse :

$\left(\begin{array}{ccc|ccc} * & * & * & 1 & 0 & 0 \\ * & * & * & 0 & 1 & 0 \\ * & * & * & 0 & 0 & 1 \end{array}\right)$

$\underbrace{\qquad\qquad}_{\text{identity matrix}}$ (under the right-hand block)

Not Inversible if : $\left(\begin{array}{ccc|c} 1 & 0 & * & \text{—} \\ 0 & 1 & * & \text{—} \\ 0 & 0 & 0 & \text{—} \end{array}\right)$

(meaning inverse is not possible)

↑ will happen

Linear Transformations

Eigenvalues → Solve in λ : $\det(A - \lambda I_n) = 0$

Eigenvector (eigen space) for λ : Solve : $(A - \lambda I_n)\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$

[If 1 of the eigen values is 0 the det = 0]

Ex. $\begin{vmatrix} -2 & 3 \\ -4 & 5 \end{vmatrix} \rightarrow \begin{vmatrix} -2-\lambda & 3 \\ -4 & 5-\lambda \end{vmatrix} = 0$

Euclidean Norm : Like Pythagorus

1. $-(2+\lambda)(5-\lambda) + \underset{\uparrow}{12} = 0$

3×-4

$-10 + 2\lambda - 5\lambda + \lambda^2 + 12 = 0$

$(\lambda - 2)(\lambda - 1) = 0$

$\boxed{\begin{matrix} \lambda_1 = 2 \\ \lambda_2 = 1 \end{matrix}}$ → Eigen Values

2. When $\lambda = 2$

$\left(\begin{matrix} -4 & 3 \\ -4 & 3 \end{matrix} \middle| \begin{matrix} 0 \\ 0 \end{matrix}\right)$

gross elimination

$\left(\begin{matrix} -4 & 3 \\ 0 & 0 \end{matrix} \middle| \begin{matrix} 0 \\ 0 \end{matrix}\right)$

$\boxed{x_2 = (\text{col.} - \text{Rank}) = 1t}$

$-4x_1 + 3x_2 = 0$

$-4x_1 = -3x_2$

$\boxed{x_1 = \frac{3}{4}t}$ $\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} \frac{3}{4} \\ 1 \end{pmatrix} \times t$

Eigen Vector

When $\lambda = 1$

$\left(\begin{matrix} -3 & 3 \\ -4 & 4 \end{matrix} \middle| \begin{matrix} 0 \\ 0 \end{matrix}\right)$

$\left(\begin{matrix} -3 & 3 \\ 0 & 0 \end{matrix} \middle| \begin{matrix} 0 \\ 0 \end{matrix}\right)$

$x_2 = t$

$x_1 = x_2 = t$

$\boxed{\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \times t}$